Exercises for Process Mining

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Notes Repository

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1 Process Discovery

1.1 Alpha Miner

1.1.1 Exercise 1

Given the following event log:

$$L_1 = [\langle a, b, c, d\rangle^3, \langle a, c, b, d\rangle^2, \langle a, e, d\rangle^1]$$

build the process model using the Alpha Miner algorithm.

Solution

1.

1.1.2 Exercise 2

Given the following event log:

$$L = \left[\langle a, b, c, d, e, f, b, d, c, e, g\rangle, \langle a, b, d, c, e, g\rangle^2, \langle a, b, c, d, e, f, b, c, d, e, f, b, d, c, e, g\rangle\right]$$

Use the Alpha Miner algorithm to discover a Petri net from the event log.

Solution:

1. First, we need to compute the footprint matrix:

	A	B	C	D	E	F	G
A	#	→	#	#	#	#	#
B		#	→	→	#	←	#
C			#	||	→	#	#
D				#	→	#	#
E					#	→	→
F						#	#
G							#

Table 1: Footprint Matrix for the Event Log L

2. Next, we identify the sets of all transitions T_L, the set of start activities T_I and the set of end activities T_O:
   - $T_L = \{a, b, c, d, e, f, g\}$
   - $T_I = \{a\}$
   - $T_O = \{g\}$

3. Then, we identify the pairs of sets of transitions that satisfy the Alpha Miner conditions. To find the first pairs, we look for causality relations in the footprint matrix:

   $$\{(\{a\},\{b\}),\ (\{b\},\{c\}),\ (\{b\},\{d\}),\ (\{f\},\{b\}),\ (\{c\},\{e\}),\ (\{d\},\{e\}),\ (\{e\},\{f\}),(\{e\},\{g\})\}$$

4. We can then group these pairs into maximal sets:

   $$\{(\{a,f\},\{b\}),\ (\{e\},\{f,g\}),\ \}$$

   And we can remove the non-maximal sets that composed them:

   $$\{(\{a\},\{b\}),\ (\{f\},\{b\}),\ (\{e\},\{f\}),\ (\{e\},\{g\})\}$$

5. Finally, we can construct the Workflow net, using one place for each of the identified pairs, and connecting them according to the causality relations:

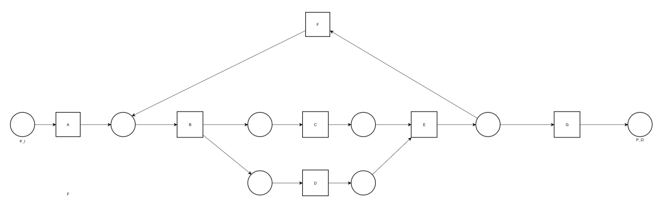

Figure 1: Petri Net Discovered using the Alpha Miner

1.1.3 Exercise 3

Given the following event log:

$$L_4 = [\langle a,c,d\rangle^45, \langle b,c,d\rangle^42, \langle a,c,e\rangle^38, \langle b,c,e\rangle^22]$$

build the process model using the Alpha Miner algorithm.

Solution

1.

1.2 Heuristic Miner

1.2.1 Exercise 1

Given the following event log:

$$L = [\langle a,e\rangle^5, \langle a,b,c,e\rangle^10, \langle a,c,b,e\rangle^10, \langle a,b,e\rangle^1, \langle a,c,e\rangle^1,$$
$$\langle a,d,e\rangle^10, \langle a,d,d,e\rangle^2, \langle a,d,d,d,e\rangle^1]$$

use the Heuristic Miner algorithm to discover a process model from the event log, with a threshold of 0.7:

Solution:

1. First, we need to compute the frequency matrix:

	A	B	C	D	E
A	0	11	11	13	5
B	0	0	10	0	11
C	0	10	0	0	11
D	0	0	0	4	13
E	0	0	0	0	0

Table 2: Frequency Matrix for the Event Log L

Each cell (x, y) contains the number of times activity x is directly followed by activity y in the event log.

2. Next, we compute the dependency matrix, by applying the dependency measure formula to each pair of activities. Remember that the dependency measure is defined as:

$$\begin{cases} \frac{|x \rightarrow y| - |y \rightarrow x|}{|x \rightarrow y| + |y \rightarrow x| + 1} & \text{if } x \neq y \\ \frac{|x \rightarrow x|}{|x \rightarrow x| + 1} & \text{otherwise} \end{cases}$$

Note that:

- Along the diagonal, we apply the second case of the formula;
- We can compute the upper triangle of the matrix, the lower triangle can be derived from it by simply swapping signs.

	A	B	C	D	E
A	0	$\frac{11}{12}$	$\frac{11}{12}$	$\frac{13}{14}$	$\frac{5}{6}$
B		0	$\frac{10-10}{21}$	0	$\frac{11}{12}$
C			0	0	$\frac{11}{12}$
D				$\frac{4}{5}$	$\frac{13}{14}$
E					0

Table 3: Dependency Matrix for the Event Log L

3. We can then build the dependecy graph, using the given threshold. We create a node for each activity in the log, and add the directed edges for each pair of activities whose dependency measure is above the threshold:

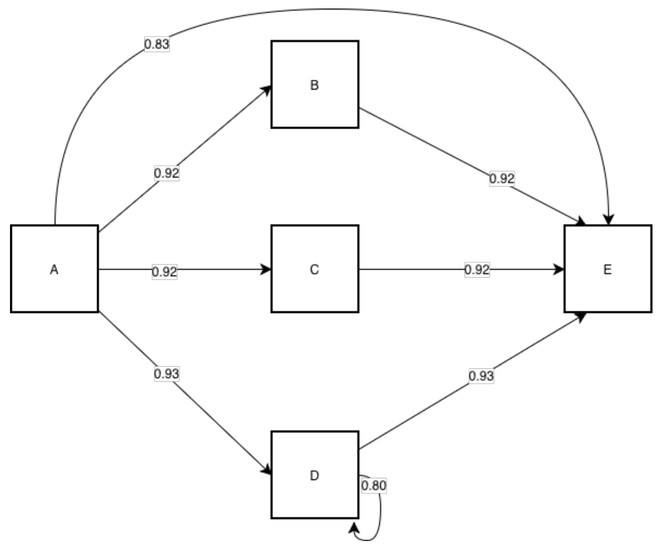

Figure 2: Dependency Graph for the Event Log L

1.2.2 Exercise 2

Given the following event log:

$$L_2 = [\langle a, b, d, e \rangle^8 0, \langle a, d, b, e \rangle^3 0, \langle a, e \rangle^2, \langle a, b, c, d, e \rangle^1 0$$

use the Heuristic Miner algorithm to discover the dependency matrix from the event log and build the dependency graph with a threshold of 0.9.

1.2.3 Exercise 3

Given the following event log:

$$L_3 = [\langle a, b, d \rangle^1 00, \langle a, c, b, d \rangle^8 0, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^4 0, \langle a, d \rangle^1$$

Compute:

- The frequency matrix;
- The dependency matrix;
- The dependency graph with a threshold of 0.9 and a threshold for direct succession of 1.